Robustifying Profile Information Propagation in Profile-Guided Optimization

Nicholas Montana

SAPIENZA
UNIVERSITÀ DI ROMA

Who Am I ?

Who Am I ?

- Cyberchallenge.it 2024 finalist

Who Am I ?

- Cyberchallenge.it 2024 finalist
- Capture-the-Flag player at the official Sapienza CTF team

Who Am I ?

- Cyberchallenge.it 2024 finalist
- Capture-the-Flag player at the official Sapienza CTF team
- Contributed to the LLVM compiler infrastructure by reporting undiscovered bugs

Performance is Critical

Performance is Critical

- Achieving optimal performance is of critical importance today.
  - A 1% performance improvement can save millions of dollars annually in large-scale deployments.

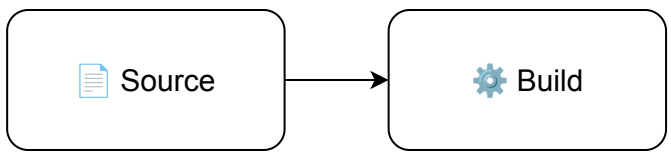
Performance is Critical

- Achieving optimal performance is of critical importance today.
  - A 1% performance improvement can save millions of dollars annually in large-scale deployments.
- **Profile-Guided Optimization** is a way to achieve great performance
  - Optimization process **tailored** to the workload of the compiled program
  - **Profile** = Execution frequencies of program regions
  - Eliminates the need to infer such information
  - Perfect for high-load applications

Profile-Guided Optimization

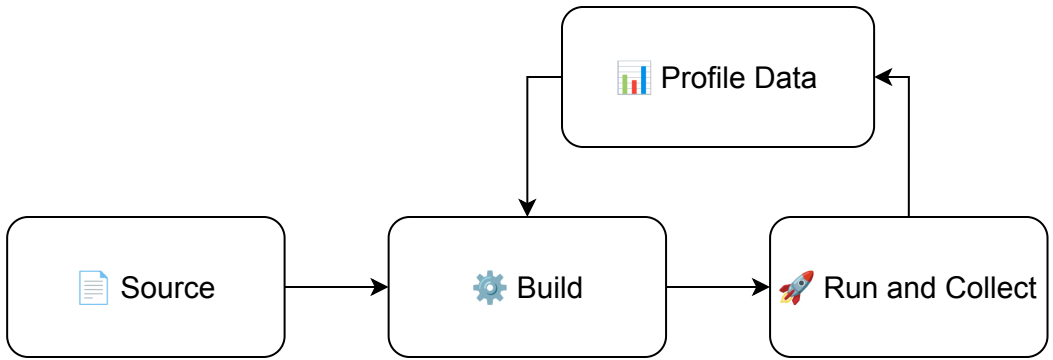
Profile-Guided Optimization

Profile-Guided Optimization

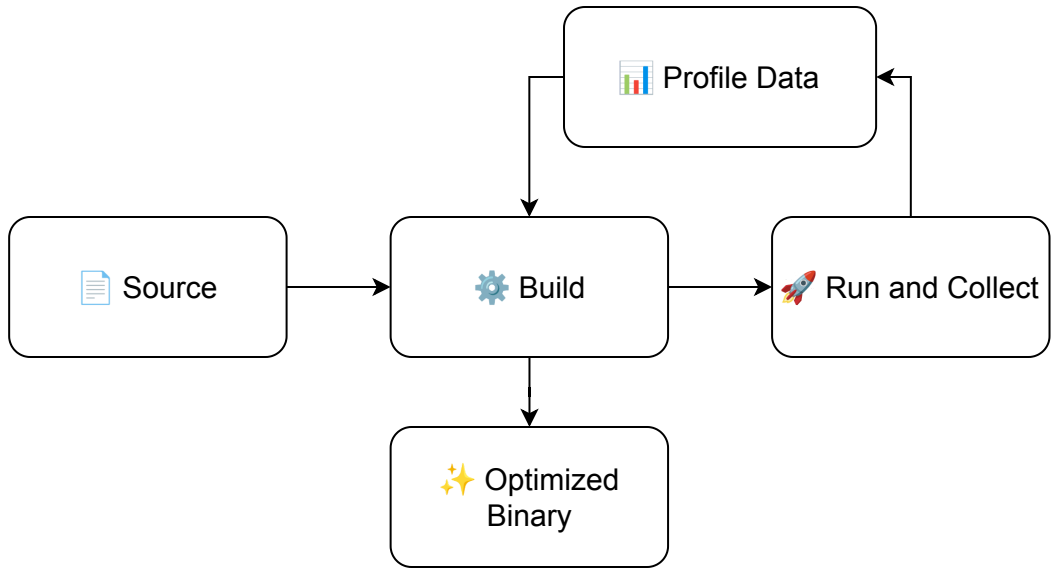


Profile-Guided Optimization

Profile-Guided Optimization



Profile-Guided Optimization



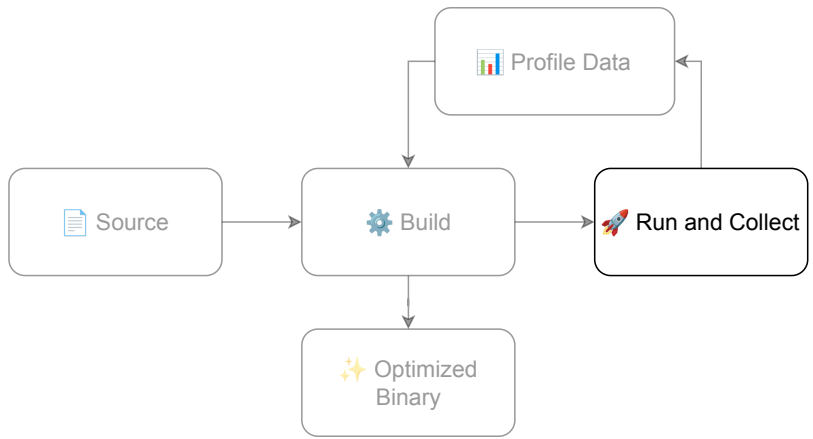
Profile Life-cycle

Profile Life-cycle

- Profiles traverse three main phases within the PGO workflow

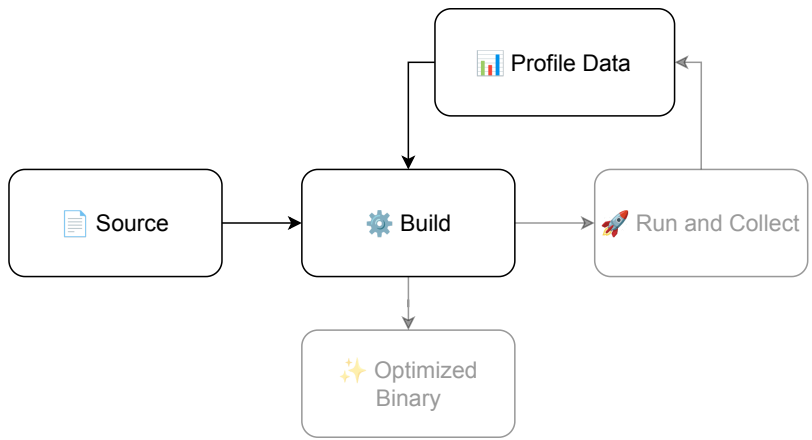
Profile Life-cycle

- Profiles traverse three main phases within the PGO workflow
  - **Collection**: The profile is collected and persisted as a file



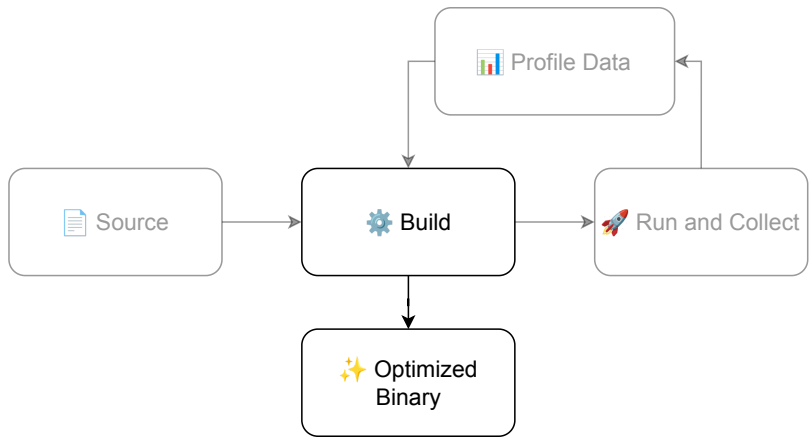
Profile Life-cycle

- Profiles traverse three main phases within the PGO workflow
  - **Collection**: The profile is collected and persisted as a file
  - **Mapping**: The profile is represented as **metadata** attached to program constructs



Profile Life-cycle

- Profiles traverse three main phases within the PGO workflow
  - **Collection**: The profile is collected and persisted as a file
  - **Mapping**: The profile is represented as **metadata** attached to program constructs
  - **Usage**: The metadata is used to guide optimizing transformations



Practical Obstacles

- For an ideal PGO application

Profile must be **complete** and **accurate** in each phase of its life-cycle

Practical Obstacles

- For an ideal PGO application

Profile must be **complete** and **accurate** in each phase of its life-cycle

- Inaccurate profiles may lead the compiler to make suboptimal optimization decisions.

Practical Obstacles

- For an ideal PGO application

Profile must be **complete** and **accurate** in each phase of its life-cycle

- Inaccurate profiles may lead the compiler to make suboptimal optimization decisions.
- In practice, each phase hides sources of inaccuracies
  - **Sampling** strategies are inaccurate by nature
  - The dynamic nature of software leads to **stale profiles**
  - Optimizing passes contain errors in **metadata propagation** logic

State of the Art

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
3. Youfeng Wu. “Accuracy of Profile Maintenance in Optimizing Compilers”. ↩
4. Profile Information Propagation Unittesting: https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595 ↩

State of the Art

- Lots of effort to solve problems in the collection and mapping phases

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
3. Youfeng Wu. “Accuracy of Profile Maintenance in Optimizing Compilers”. ↩
4. Profile Information Propagation Unittesting: https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595 ↩

State of the Art

- Lots of effort to solve problems in the collection and mapping phases
  - [1] Proposes a rectification algorithm to rectify sampled profiles

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
3. Youfeng Wu. “Accuracy of Profile Maintenance in Optimizing Compilers”. ↩
4. Profile Information Propagation Unittesting: https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595 ↩

State of the Art

- Lots of effort to solve problems in the collection and mapping phases
  - [1] Proposes a rectification algorithm to rectify sampled profiles
  - [2] Proposes an algorithm to adapt stale profiles to newer program versions

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
3. Youfeng Wu. “Accuracy of Profile Maintenance in Optimizing Compilers”. ↩
4. Profile Information Propagation Unittesting: https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595 ↩

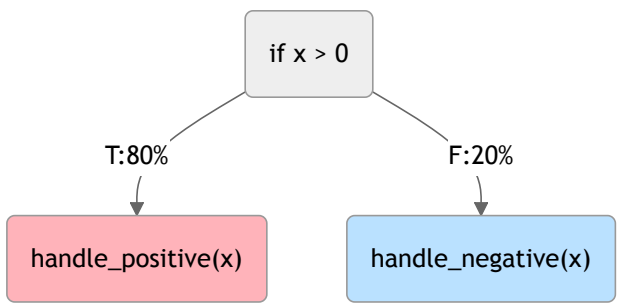
State of the Art

- Lots of effort to solve problems in the collection and mapping phases
  - [1] Proposes a rectification algorithm to rectify sampled profiles
  - [2] Proposes an algorithm to adapt stale profiles to newer program versions
  - [3], [4] only partially address failures in metadata propagation

1. Wenlei He, Julián Mestre, Sergey Pupyrev, Lei Wang, and Hongtao Yu. “Profile inference revisited”. ↩
2. Amir Ayupov, Maksim Panchenko, and Sergey Pupyrev. “Stale Profile Matching”. ↩
3. Youfeng Wu. “Accuracy of Profile Maintenance in Optimizing Compilers”. ↩
4. Profile Information Propagation Unittesting: https://discourse.llvm.org/t/rfc-profile-information-propagation-unittesting/73595 ↩

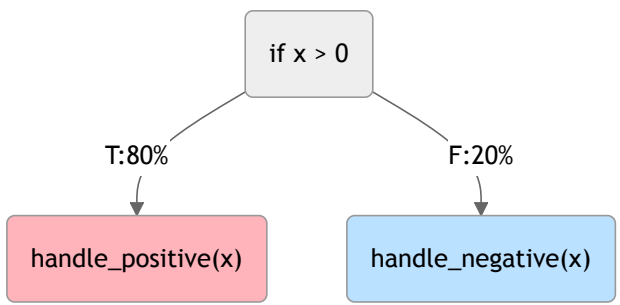
Example of Profile Mishandling

```
// Before pass
if (x > 0) { // Then branch taken 80 times
    handle_positive(x) 🔥
} else { // Else branch taken 20 times
    handle_negative(x) ❄️
}
```

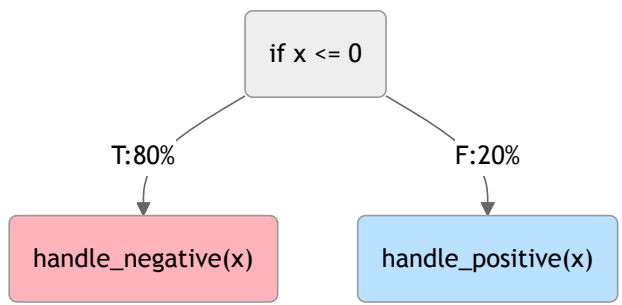


Example of Profile Mishandling

```
// Before pass
if (x > 0) { // Then branch taken 80 times
    handle_positive(x) 🔥
} else { // Else branch taken 20 times
    handle_negative(x) ❄️
}
```




```
// After pass
if (x ⩽ 0) { // Then branch taken 80 times
    handle_negative(x) 🔥
} else { // Else branch taken 20 times
    handle_positive(x) ❄️
}
```



Profile Mishandling Research Gap

Profile Mishandling Research Gap

- Recently discovered **regressions** could be attributed to profile mishandling

Profile Mishandling Research Gap

- Recently discovered **regressions** could be attributed to profile mishandling
- Profile mishandling **nullifies** solutions aiming at correcting profiles in earlier stages

Profile Mishandling Research Gap

- Recently discovered **regressions** could be attributed to profile mishandling
- Profile mishandling **nullifies** solutions aiming at correcting profiles in earlier stages
- **Non-trivial** solution
  - What does it mean to correctly propagate profiles?
  - No immediate correlation between before and after code
  - **Interaction** between passes needs to be taken into account
  - **Static checks** on control-flow are not enough!
  - **No dedicated tools** to validate propagation logic

Profile Mishandling Research Gap

- Recently discovered **regressions** could be attributed to profile mishandling
- Profile mishandling **nullifies** solutions aiming at correcting profiles in earlier stages
- **Non-trivial** solution
  - What does it mean to correctly propagate profiles?
  - No immediate correlation between before and after code
  - **Interaction** between passes needs to be taken into account
  - **Static checks** on control-flow are not enough!
  - **No dedicated tools** to validate propagation logic

Profile information is **transformed** together with the program, but unlike the program itself, its correctness cannot be directly observed.

Research Proposal

Can profile propagation accuracy be assessed systematically?

Research Proposal

Can profile propagation accuracy be assessed systematically?

- A methodological and practical framework to
  - Validate profile information propagation using **random** C programs
  - Improve the PGO logic coverage of the tested compiler via **coverage-guided** testing

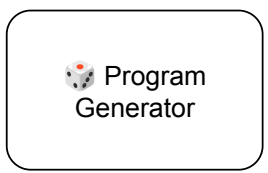
Stress-Test approach

Stress-Test approach

- Stress-testing profile propagation logic with randomly generated programs

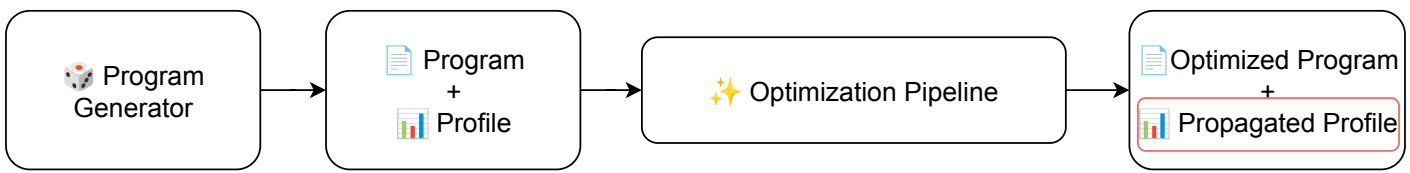
Stress-Test approach

- Stress-testing profile propagation logic with randomly generated programs
  - Uses **off-the-shelf** random program generators



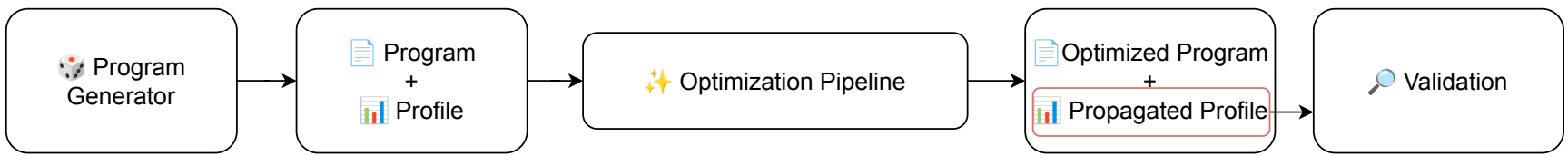
Stress-Test approach

- Stress-testing profile propagation logic with randomly generated programs
  - Uses **off-the-shelf** random program generators
  - Random program are optimized to get the **propagated profile**



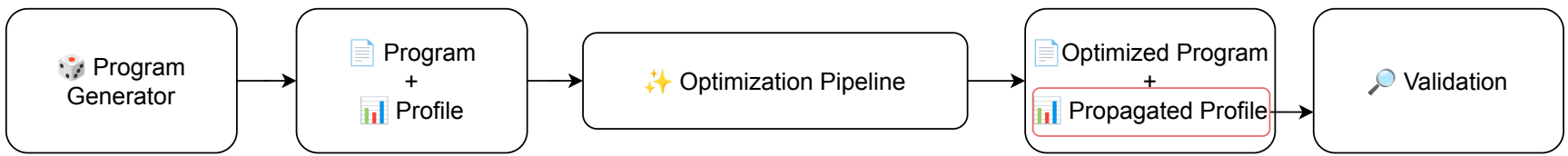
Stress-Test approach

- Stress-testing profile propagation logic with randomly generated programs
  - Uses **off-the-shelf** random program generators
  - Random program are optimized to get the **propagated profile**
  - Profile **validation** to spot profile mishandling



Stress-Test approach

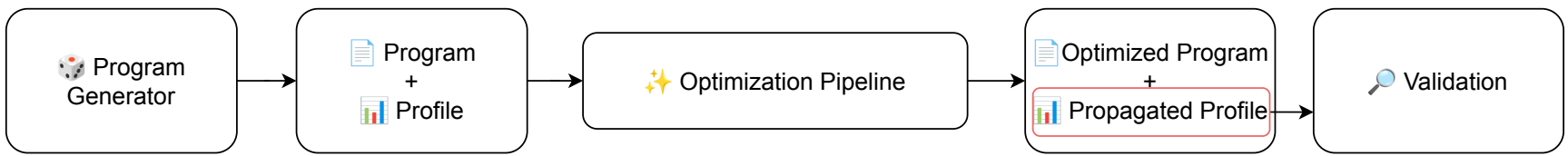
- Stress-testing profile propagation logic with randomly generated programs
  - Uses **off-the-shelf** random program generators
  - Random program are optimized to get the **propagated profile**
  - Profile **validation** to spot profile mishandling



- Automated **bug triaging** to classify the large number of issues

Stress-Test approach

- Stress-testing profile propagation logic with randomly generated programs
  - Uses **off-the-shelf** random program generators
  - Random program are optimized to get the **propagated profile**
  - Profile **validation** to spot profile mishandling



- Automated **bug triaging** to classify the large number of issues
- Efficacy is bounded by the **complexity** achievable by program generators

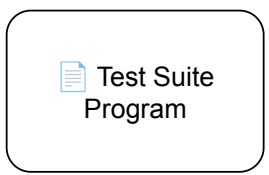
Coverage-Guided Testing

Coverage-Guided Testing

- A deeper analysis approach, which uses

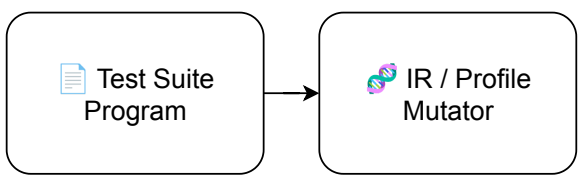
Coverage-Guided Testing

- A deeper analysis approach, which uses
  - **Test-suite** programs as a foundation



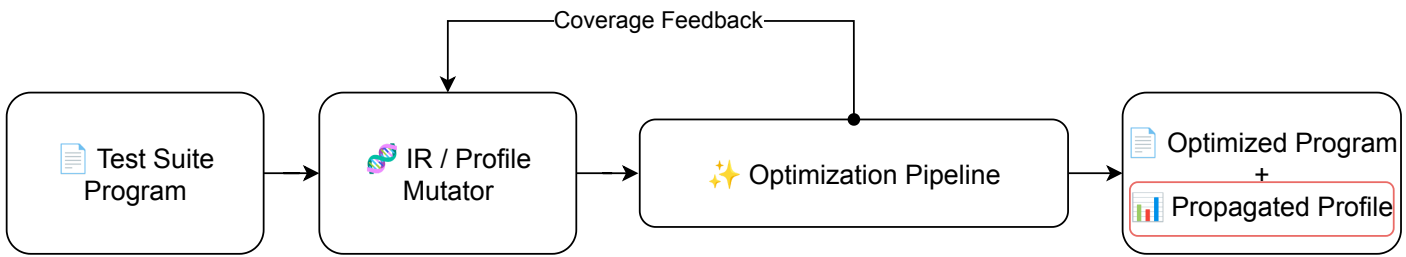
Coverage-Guided Testing

- A deeper analysis approach, which uses
  - **Test-suite** programs as a foundation
  - Classic code mutations and **novel** profile mutations strategies



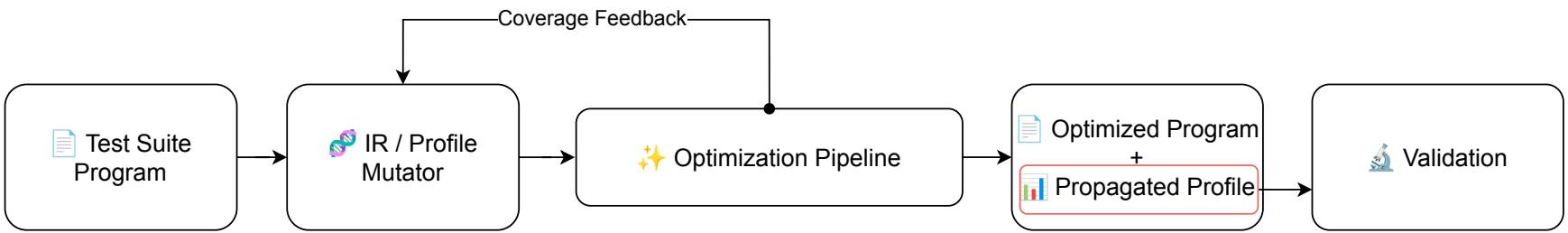
Coverage-Guided Testing

- A deeper analysis approach, which uses
  - **Test-suite** programs as a foundation
  - Classic code mutations and **novel** profile mutations strategies
  - **Feedback** mechanism that instantiates a coverage metric to guide mutations



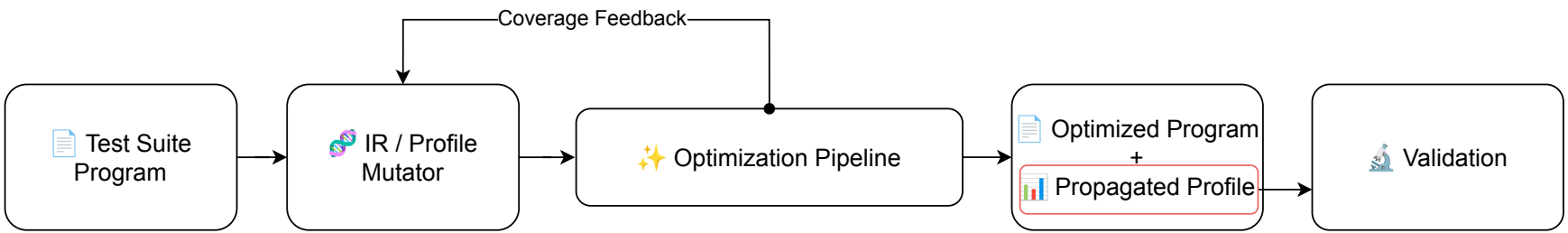
Coverage-Guided Testing

- A deeper analysis approach, which uses
  - **Test-suite** programs as a foundation
  - Classic code mutations and **novel** profile mutations strategies
  - **Feedback** mechanism that instantiates a coverage metric to guide mutations
  - Profile **validation** to spot profile mishandling



Coverage-Guided Testing

- A deeper analysis approach, which uses
  - **Test-suite** programs as a foundation
  - Classic code mutations and **novel** profile mutations strategies
  - **Feedback** mechanism that instantiates a coverage metric to guide mutations
  - Profile **validation** to spot profile mishandling



- Exposes **untested** regions of the compiler to uncover deeper issues

Evaluation of the Proposed Directions

Evaluation of the Proposed Directions

- **LLVM** compiler infrastructure as evaluation target

Evaluation of the Proposed Directions

- **LLVM** compiler infrastructure as evaluation target
- Can the proposed methodologies:

Evaluation of the Proposed Directions

- **LLVM** compiler infrastructure as evaluation target
- Can the proposed methodologies:
  - Find new **bugs**? If so, how many?

Evaluation of the Proposed Directions

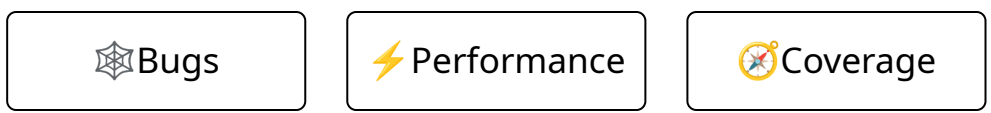
- **LLVM** compiler infrastructure as evaluation target
- Can the proposed methodologies:
  - Find new **bugs**? If so, how many?
  - **Improve the performance** of the generated binaries?

🕸️Bugs

⚡Performance

Evaluation of the Proposed Directions

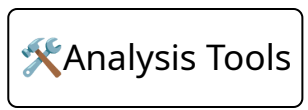
- **LLVM** compiler infrastructure as evaluation target
- Can the proposed methodologies:
  - Find new **bugs**? If so, how many?
  - **Improve the performance** of the generated binaries?
  - Improve optimization pass coverage within LLVM?



Impacts and Benefits

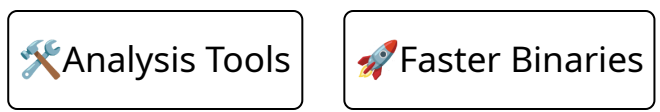
Impacts and Benefits

- **Analysis tools** for compiler developers, to assess their PGO implementations



Impacts and Benefits

- **Analysis tools** for compiler developers, to assess their PGO implementations
- **Faster** PGO binaries for the user



Impacts and Benefits

- **Analysis tools** for compiler developers, to assess their PGO implementations
- **Faster** PGO binaries for the user
- **Energy savings** due to optimal binaries

Thanks for the attention!

Questions?